Module 5: Assignment

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IFT 372: Wireless Fundamentals

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Questions

1.

a. Explain the concept of 5G network slicing.

5G network slicing allows for the existence of virtual networks on top of an existing physical spectrum band. Segmenting the network with network slicing allows the network to be divided by users or use cases.

b. Identify any benefits for high-speed Internet access.

High-speed internet access permits quite a few benefits for users. High-speed internet access enables high-bitrate activities such as 4K video streaming. It provides a generally better customer experience as annoyances like video buffering become less likely. Latency-sensitive activities such as mobile multiplayer gaming would also greatly benefit from this.

2. Describe the basic concept of the 5G Non-Standalone (NSA) architecture.

The basic concept of 5G NSA architecture is the 5G network is built on top of existing the 4G core network and certain functions such as authentication and session management are carried out over the 4G network. This leaves more resources free on the 5G network to handle higher-speed traffic.

3. Explain how Dynamic Spectrum Sharing (DSS) can be used for spectrum refarming.

Dynamic Spectrum Sharing is the dynamic software that looks at user type and traffic load to handle spectrum allocation flexibly. This is the mechanism that allows for the efficient reallocation of spectrum resources without disrupting services. It also enables traffic to have a smooth shift from 4G to 5G networks.

4. **Explain the concept of the service oriented architecture for the 5G core network.**

Service-oriented architecture in 5G presents the control plane as separated into services that provide network functions such as authentication and session management. It presents these services closer to a client/server model rather than the interface model used in 4G. This architecture also allows for automated NF service discovery.

5. **Explain the differences between the TDD and FDD 5G air interface.**

Frequency Division Duplex of 5G is the transmission mode that uses two separate frequency bands; one for uplink and the other for downlink. Time Division Duplex is the transmission mode of having both uplink and downlink on the same frequency band where data is transmitted in only one direction at a time within its own time slots.

6. **What is a Bandwidth Part (BWP)?**

A Bandwidth Part in 5G is defined as a subsection of the total bandwidth that can be allocated to a certain user or users (arminderkaur192, 2024). It enables dynamically adjusting the amount of bandwidth depending on the needs of the user or users and the availability of network resources in order to have the most efficient use of the spectrum.

7. What is a CORESET?

A CORESET or Control Resource Set is a configurable space on the NR Downlink Resource Grid that is used to transmit essential downlink information such as PDCCH/DCI. CORESET plays an important role in effective resource allocation and the transmission of signaling in the control plane.

8. What is a split-bearer?

A split-bearer is an essential component of Dual Connectivity in 5G. A split-bearer allows the UE to send data down two paths. A path to a 5G gNodeB and another path to a 4G eNodeB. In this configuration, control information can be transmitted via the path to the eNodeB and user data can be transmitted through the faster gNodeB path. This allows for more efficient utilization of 4G and 5G technologies by using the capabilities of both.

9. What is the difference between Registration and Session Management?

Registration handles the establishment of connections between the UE and the network and is worried about the location and identity of the UE. Session Management handles control data and tries to maintain good QoS.

10. What is the difference between the RRC-Idle and the RRC-Inactive state?

RRC-Idle state is where the UE has no active connections to the network. It is used when no data is being transmitted to the network. The UE must go through a connection process when leaving this state. RRC-Inactive state is where the UE is still connected to the network but no data is being transmitted but it can easily start transmitting to the network when needed.

11. What is the purpose of the container in the 5G core?

Containers can be run to provide 5G network function services in a portable, scalable, and platform-agnostic way via software such as Docker. Network function services in service-oriented architecture can be scaled to network demand with container orchestration software such as Kubernetes to ensure high availability and fault tolerance at a service level.

12. Why are two UE transmitters required for a 5Goption 3 split-bearer connection?

Two UE transmitters are required because split-bearer connections separate the user plane and the control plane and send them down two routes; one to a gNodeB and the other to an eNodeB. The UE is leveraging two different connections via two different technologies at the same time so two UE transmitters are required.

13. Why can the 4G and 5G part of a 5G NSA bearer be handed over independently from each other?

5G NSA already uses both 5G and 4G networks through Dual Connectivity. The 4G and 5G parts of the NSA bearer can be handed over separately because the UE is connected to both 4G and 5G networks simultaneously.

References

arminderkaur192. (2024, September 16). *What is 5g nr bandwidth part(bwp)*. TELCOMA Training & Certifications. https://telcomatraining.com/what-is-5g-nr-bandwidth-partbwp/